

MATHEMATICS

9709/43

Paper 4 Mechanics 1 (M1)

October/November 2015

1 hour 15 minutes

Additional Materials: Answer Booklet/Paper
Graph Paper
List of Formulae (MF9)

READ THESE INSTRUCTIONS FIRST

If you have been given an Answer Booklet, follow the instructions on the front cover of the Booklet.

Write your Centre number, candidate number and name on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

Where a numerical value for the acceleration due to gravity is needed, use 10 m s^{-2} .

The use of an electronic calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

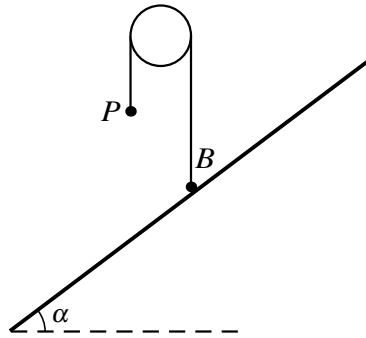
The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 50.

Questions carrying smaller numbers of marks are printed earlier in the paper, and questions carrying larger numbers of marks later in the paper.

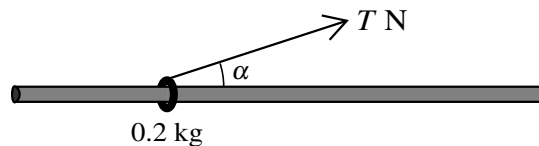
This document consists of 4 printed pages.

1



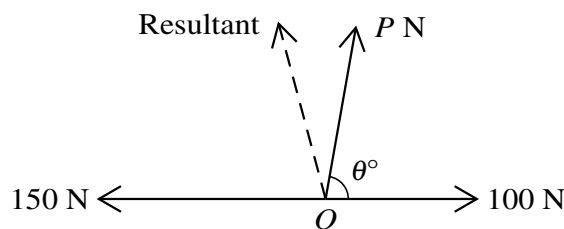
A small ball B of mass 4 kg is attached to one end of a light inextensible string. A particle P of mass 3 kg is attached to the other end of the string. The string passes over a fixed smooth pulley. The system is in equilibrium with the string taut and its straight parts vertical. B is at rest on a rough plane inclined to the horizontal at an angle of α , where $\cos \alpha = 0.8$ (see diagram). State the tension in the string and find the normal component of the contact force exerted on B by the plane. [3]

2



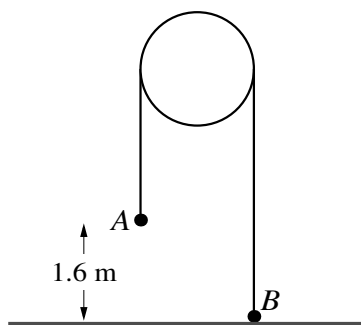
A ring of mass 0.2 kg is threaded on a fixed rough horizontal rod and a light inextensible string is attached to the ring at an angle α above the horizontal, where $\cos \alpha = 0.96$. The ring is in limiting equilibrium with the tension in the string $T\text{ N}$ (see diagram). Given that the coefficient of friction between the ring and the rod is 0.25 , find the value of T . [5]

3



Three horizontal forces of magnitudes 150 N , 100 N and $P\text{ N}$ have directions as shown in the diagram. The resultant of the three forces is shown by the broken line in the diagram. This resultant has magnitude 120 N and makes an angle 75° with the 150 N force. Find the values of P and θ . [7]

4



Particles A and B , of masses 0.35 kg and 0.15 kg respectively, are attached to the ends of a light inextensible string which passes over a fixed smooth pulley. The system is at rest with B held on the horizontal floor, the string taut and its straight parts vertical. A is at a height of 1.6 m above the floor (see diagram). B is released and the system begins to move; B does not reach the pulley. Find

- (i) the acceleration of the particles and the tension in the string before A reaches the floor, [4]
- (ii) the greatest height above the floor reached by B . [3]

- 5 A cyclist and his bicycle have a total mass of 90 kg . The cyclist starts to move with speed 3 m s^{-1} from the top of a straight hill, of length 500 m , which is inclined at an angle of $\sin^{-1} 0.05$ to the horizontal. The cyclist moves with constant acceleration until he reaches the bottom of the hill with speed 5 m s^{-1} . The cyclist generates 420 W of power while moving down the hill. The resistance to the motion of the cyclist and his bicycle, $R\text{ N}$, and the cyclist's speed, $v\text{ m s}^{-1}$, both vary.

- (i) Show that $R = \frac{420}{v} + 43.56$. [5]
- (ii) Find the cyclist's speed at the mid-point of the hill. Hence find the decrease in the value of R when the cyclist moves from the top of the hill to the mid-point of the hill, and when the cyclist moves from the mid-point of the hill to the bottom of the hill. [3]

- 6 A particle P starts from rest at a point O of a straight line and moves along the line. The displacement of the particle at time $t\text{ s}$ after leaving O is $x\text{ m}$, where

$$x = 0.08t^2 - 0.0002t^3.$$

- (i) Find the value of t when P returns to O and find the speed of P as it passes through O on its return. [4]
- (ii) For the motion of P until the instant it returns to O , find
 - (a) the total distance travelled, [3]
 - (b) the average speed. [2]

[Question 7 is printed on the next page.]

- 7 A straight hill AB has length 400 m with A at the top and B at the bottom and is inclined at an angle of 4° to the horizontal. A straight horizontal road BC has length 750 m. A car of mass 1250 kg has a speed of 5 m s^{-1} at A when starting to move down the hill. While moving down the hill the resistance to the motion of the car is 2000 N and the driving force is constant. The speed of the car on reaching B is 8 m s^{-1} .

(i) By using work and energy, find the driving force of the car. [5]

On reaching B the car moves along the road BC . The driving force is constant and twice that when the car was on the hill. The resistance to the motion of the car continues to be 2000 N. Find

(ii) the acceleration of the car while moving from B to C , [3]

(iii) the power of the car's engine as the car reaches C . [3]

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